Lesson plan · for the teacher

Leading data science at school: unit check

Data science · Level 4 · Unit check

Time	Materials
40 min	Board, chalk, paper. No devices needed.

Learning goals

- Samples and margins
- Finding groups
- Fair models
- Deciding with data

Activity: Team quiz

1. Teams (2 min): Make teams of four and give each team a sheet for its answers.
2. Quiz (20 min): Read each question aloud; each team agrees on one answer and writes it down.
3. Explain (10 min): Reveal each answer and ask a team to explain it with an example from the unit.
4. Wrap-up (8 min): Work through the question most teams got wrong, together on the board.

Discussion

- Which idea will you bring to staff first?
- Which school tool will you audit?
- Which hunch will you test?

Exit ticket

- One trial I will propose this term, and its comparison group.

Try it in class tomorrow

Bring one trial plan to your next staff meeting and agree on a date to share its results.

Worksheet · for pupils

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Name _____ Class _____ Date _____

1 **Two fair random samples always give the same average.**

◯ True ◯ False

2 **A sample is four times bigger. What happens to the margin of error?**

◯ It halves ◯ It quarters ◯ It doubles

3 **Put the k-means steps in order.**

☐ Repeat

☐ Move centres to the mean

☐ Join the nearest centre

☐ Place k centres

4 **A survey only asked pupils at the after-school club. What is the main problem?**

◯ It is not random, so some pupils are missing ◯ It is too small ◯ It used paper

5 **A cluster's name is a fact found by the computer.**

◯ True ◯ False